

Triad strict-CI beats Lange at 5 of 8 $d \geq 7$ ops points;
loses to PM at $d=9$ $p=0.015$ (above threshold)

d=3, p=0.005	overlap	—	overlap	WIN +7 bp
d=3, p=0.007	overlap	—	overlap	WIN +21 bp
d=3, p=0.01	overlap	—	overlap	WIN +36 bp
d=3, p=0.015	overlap	—	overlap	WIN +58 bp
d=5, p=0.005	loss -7 bp	—	overlap	WIN +28 bp
d=5, p=0.007	loss -22 bp	—	overlap	WIN +70 bp
d=5, p=0.01	loss -57 bp	—	overlap	WIN +126 bp
d=5, p=0.015	loss -90 bp	—	WIN +7 bp	WIN +173 bp
d=7, p=0.005	overlap	overlap	WIN +0 bp	WIN +25 bp
d=7, p=0.007	WIN +26 bp	WIN +5 bp	WIN +37 bp	WIN +78 bp
d=7, p=0.01	WIN +122 bp	WIN +55 bp	WIN +171 bp	WIN +125 bp
d=7, p=0.015	WIN +231 bp	WIN +27 bp	WIN +377 bp	WIN +74 bp
d=9, p=0.005	loss -12 bp	—	overlap	WIN +17 bp
d=9, p=0.007	loss -48 bp	—	WIN +15 bp	WIN +68 bp
d=9, p=0.01	loss -155 bp	—	WIN +183 bp	WIN +98 bp
d=9, p=0.015	loss -101 bp	—	WIN +336 bp	loss -56 bp

PFWL3S vs Lange-pub PFWL3S vs Lange-FT Triad vs Lange-pub Triad vs PyMatching

strict-CI WIN (95% non-overlap) overlap (statistical tie) strict-CI loss

Wilson 95% CIs at 100K shots / point. WIN = row-decoder strictly beats column-decoder. Numbers show CI-gap magnitude in basis points. The single 'loss' cell at $d=9$ $p=0.015$ in the right column reflects that the $d=9$ surface code is above its pseudo-threshold at $p=0.015$, where PM's combinatorial structure is provably near-optimal.